BIG IDEAS MATH • GRADE 7 • CHAPTER 3

Expressions & Equations: Two-Step Equations & Balance Stage

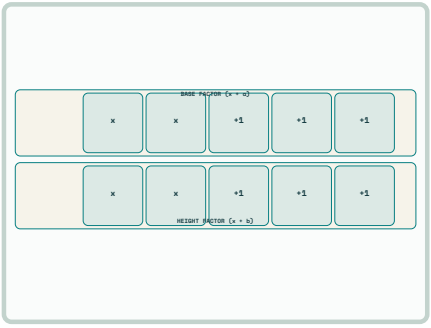
Tactile Concrete-Pictorial-Abstract (CPA) Station Guide • CCSS.MATH.7.EE.B.4

Student Name: _____ Date: _____ **Model: SC-32** **Check Score: ___ / 4**

STUDENT SELF-CHECK RULE (NO CHEATING):
Build with blocks and solve on paper first! Use your SC-32 calculator ONLY to verify answers on the 4-line LCD screen.

STATION LAB TUTORIAL: BENCH SCHEMATIC & CPA PROGRESSION

1. 2D Bench Blueprint



Kit: Algebra Stage

2. Desk Setup (Concrete -> Pictorial)

Physical Bench Setup:
Place two x-rods and 5 unit tiles on left tray; place 17 units on right tray. Undo addition first, then divide.

Sketch Model on Paper:
Draw two-pan balance scale: remove 5 from both pans, then divide remaining 12 into 2 equal piles.

3. Step-by-Step LCD Stack Trace (SC-32)

MODE: 123 Base-10 Arithmetic

1.	28	X: 28	Target total 28 into
2.	ENTER	Y: 28 | X: 28	Stack lift
3.	7	Y: 28 | X: 7	Constant term 7 into
4.	-	X: 21	Undo addition: 28 -

EXPECTED LCD X: x = 7

1 **Solve 3x + 8 = 29 on the stack.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer:

☐ Verified on SC-32

2 **Solve 5x - 7 = 28 on the stack.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer:

☐ Verified on SC-32

3 **Solve 4x + 12 = 48 on the stack.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer:

☐ Verified on SC-32

4 **Solve (x + 5) / 4 = 7 on the stack.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer:

☐ Verified on SC-32

Station Diagnostic & Solution Key: Chapter 3

Expressions & Equations: Two-Step Equations & Balance Stage • Manipulative Setup, Diagnostic Rubric & Verification Flow

MANIPULATIVE HARDWARE STATION INVENTORY & 3D PRINT PREPARATION

Station Kit: Tactile Algebra Stage & Tile Weights

CAD Blueprint: algebra_stage

Target Device: SC-32 Scientific

CAD / STL Source File:

Education/build/scad/algebra_stage.scad

Standard: Gridfinity 42mm Pitch / 0.6mm Inset Markings

Recommended Filaments:

Oceanic Teal Stage Frame & White Algebra Tiles (PLA)

#	Math Problem Prompt	Expected Math Answer	Self-Check Keystrokes & RPN Stack
1	**Solve 3x + 8 = 29 on the stack.** Standard: CCSS.MATH.7.EE.B.4	x = 7	[2][9] [ENTER] [8] [-] [3] [÷]
2	**Solve 5x - 7 = 28 on the stack.** Standard: CCSS.MATH.7.EE.B.4	x = 7	[2][8] [ENTER] [7] [+] [5] [÷]
3	**Solve 4x + 12 = 48 on the stack.** Standard: CCSS.MATH.7.EE.B.4	x = 9	[4][8] [ENTER] [1][2] [-] [4] [÷]
4	**Solve (x + 5) / 4 = 7 on the stack.** Standard: CCSS.MATH.7.EE.B.4	x = 23	[7] [ENTER] [4] [×] [5] [-]

TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

Key Misconception to Target During Station Observation:

Students often divide before subtracting the constant term.

How the Manipulatives & Calculator Foster Self-Correction:

Students solve with pencil and physical blocks first. When they self-check on the SC-32, any discrepancy between their written answer and the calculator's 4-line LCD immediately prompts them to re-inspect their physical bench model or regrouping, turning errors into productive struggle without teacher intervention.

Station Facilitation & Pacing Tip:

Keep 3D printed trays docked on standard Gridfinity baseplates to prevent loose parts on student desks. Encourage students to articulate their CPA steps: name the block representation (Concrete), point to their drawn tape/bond model (Pictorial), and verify register values (Abstract).

CLASSROOM SCORING & STATION MASTERY RUBRIC

Exemplary (18–20 pts):	Complete, accurate manipulative sketches; exact math answers; independently verified self-check on LCD stack.
Proficient (14–17 pts):	Correct math answers; minor sketch omissions; verified on calculator with minimal teacher prompting.
Developing (10–13 pts):	Understands problem concept but made computational slips; successfully self-corrected using calculator LCD.
Intervention (<10 pts):	Relied on calculator before solving on paper, or struggled with underlying physical manipulative models.